Venkata Saratchandra Patnaik Markonda Patnaikuni

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EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4) **December 2025**
Arizona State University, Tempe, AZ.
Bachelor's of Technology in Computer Science (CGPA: 9.11/10) **July 2018 - May 2022**
Arizona State University, Tempe, AZ.
Relevant Coursework: Foundations of Algorithms, Machine Learning, Data Processing at Scale, Cloud Computing, Human-Computer Interaction, Applied Cryptography, Artificial Intelligence, Statistical Foundations for AI, Introduction to Data Science

TECHNICAL SKILLS

Programming Languages: Python, R, Java, Kotlin, Go, JavaScript, TypeScript, C, C#, Linux/Shell Scripting
Web & Mobile Development: React.js, Node.js, Express.js, HTML, CSS, Bootstrap, Android Studio, Flask, PyTorch
Databases: MySQL, PostgreSQL, Firebase, MariaDB/ MongoDB
DevOps & Cloud: AWS (S3, EKS, EC2, Lambda, Load Balancer), Docker, Kubernetes, Jenkins, REST APIs
CI/CD & Version Control tools: GitHub, GitOps, ArgoCD, Jira, Jenkins

PROFESSIONAL EXPERIENCE

Software Engineer: Amagi Media Labs, Bengaluru, India **August 2022 – November 2023**

- **Solution Architecture & Automation:** Involved in **cloud-native** infrastructure initiatives by automating onboarding and deployment workflows using **Terraform**, reducing manual configuration time by 90% and improving operational efficiency.
- Managed and deployed **15+ microservices** on **AWS EKS** using **Docker** and **Kubernetes**, enhancing system scalability and cutting release cycle times by 30%
- Implemented **CI/CD pipelines** with **ArgoCD** and **GitOps**, integrating **Python**-based **validation** tools that reduced content delivery failures by 95% and improved system rollback reliability.

Web Development Intern: Blueplanet Solutions Inc., India **April 2021 - June 2021**

- Engineered a **MySQL database** with **stored procedures** for 'Campus Club,' accommodating **1,000+ student profiles** and enabling efficient data retrieval, reducing query response times by **60%,** and enhancing user experience.
- Enhanced a responsive front-end student search feature using **HTML, CSS, and JavaScript**, streamlining the student search process, which **decreased average search time by 60%**.
- Evaluated web application logs using data analysis techniques, pinpointing and **resolving memory leaks**, which improved **application stability** by **30%** and reduced server **CPU utilization** by **10%.**

PROJECTS

Plethora - A Fest Management System (Full Stack)

- Built with **React.js, Node.js, HTML, CSS, JavaScript, MariaDB, and MySQL** for event discovery, registration, and authentication, improving participation by **40%**. **Enhanced** MariaDB database **performance** through in-depth query analysis and index optimization, decreasing real-time **query response time** by **50%** and reducing **server load** by **15%**.

Expense Tracker (Android Studio)

- Developed a mobile expense tracker application using **Kotlin, Java, and Firebase**; enabled real-time budget monitoring and optimized **Firebase sync**, reducing latency by **33%** for a smoother user experience.

Handling Imbalanced Data on Bank Churn Prediction (Machine Learning & Data Analytics)

- Developed a machine learning model to predict customer churn using **Python**, addressing imbalanced datasets with **SMOTE** and **undersampling**, improving model accuracy by 15%.
- Performed data cleaning and visualization with **Pandas** and **Matplotlib**, extracting key features for classification that were relevant to multimodal data processing in vision research